SCSJ 3343 – Software Quality Assurance

Software Test Management Using TestLink

Version 1.9.17



Dr. Wan Mohd Nasir Wan Kadir
School of Computing
Faculty of Engineering
Universiti Teknologi Malaysia

Premier Digital Tech University™

Agenda

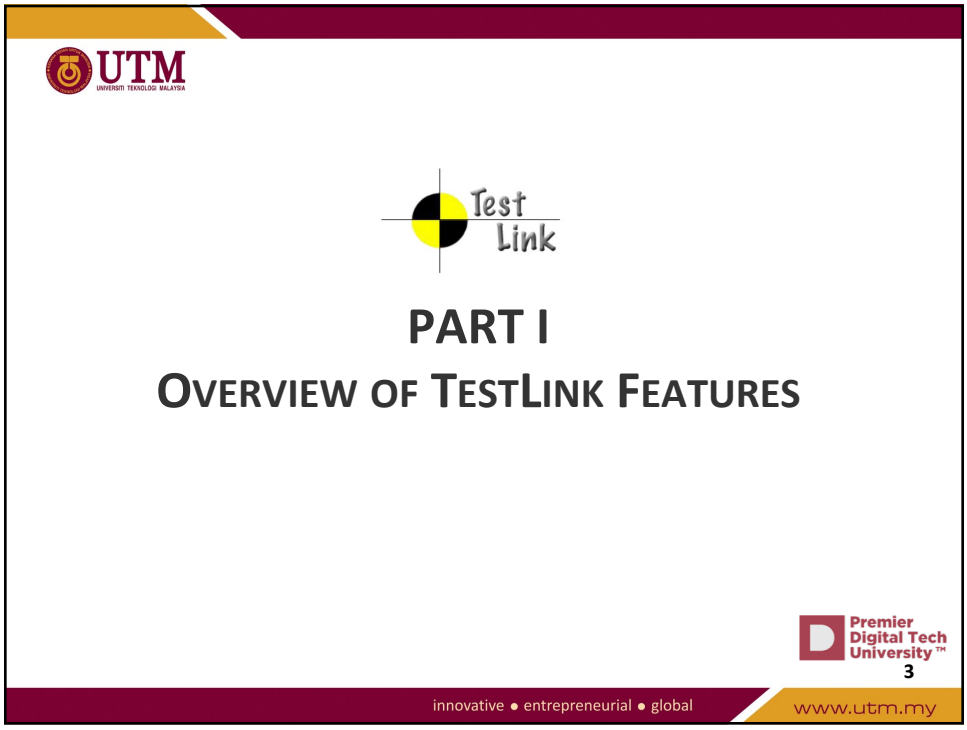
PART I - OVERVIEW OF TESTLINK FEATURES

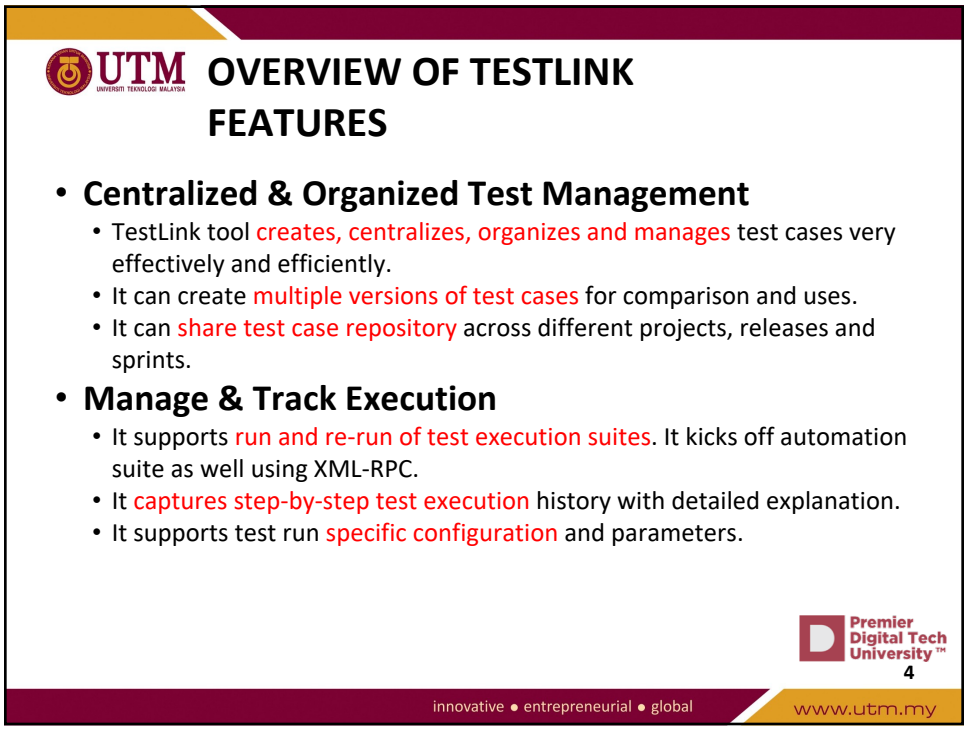
PART II - CREATING A PROJECT, TEST SUITE, AND TEST CASES

PART III - CREATING A TEST PLAN, SPECIFY BUILD, AND ADDING THE TEST CASES TO THE TEST PLAN

PART IV - ASSIGNING TEST CASE EXECUTION, EXECUTING TEST CASES, AND GENERATING TEST REPORTS

Premier Digital Tech University™


UTM
Test Link
PART I
OVERVIEW OF TESTLINK FEATURES
Premier Digital Tech University™
3
innovative ● entrepreneurial ● global
www.utm.my


UTM
OVERVIEW OF TESTLINK FEATURES
• Centralized & Organized Test Management
• TestLink tool creates, centralizes, organizes and manages test cases very effectively and efficiently.
• It can create multiple versions of test cases for comparison and uses.
• It can share test case repository across different projects, releases and sprints.
• Manage & Track Execution
• It supports run and re-run of test execution suites. It kicks off automation suite as well using XML-RPC.
• It captures step-by-step test execution history with detailed explanation.
• It supports test run specific configuration and parameters.
Premier Digital Tech University™
4
innovative ● entrepreneurial ● global
www.utm.my

OVERVIEW OF TESTLINK FEATURES

Other Features:

- Assigning Keywords (we may form a group of Test Cases for Regression tests)
- Specifying Requirements (we may bind them with Test Cases in the many-to-many relation and see if our Test Cases cover our requirements)
- Events log (you can see here the history of all the changes)

Premier Digital Tech University™

5

TESTLINK BENEFITS

- We have all the documents structured and *organized*
- We solve the problem of version management *We can control the testing process (events log + different type of reports)*
- We can see if all the requirements are covered with *Test Cases*
- We can select Test Cases for different type of *testing (Regression, Acceptance, Smoke)*
- We can see the results of testing in a very clear and *easy-to-use form*
- We can track in real time the status of the testing *process*

Premier Digital Tech University™

6

innovative ● entrepreneurial ● global
www.utm.my

WORKING WITH TESTLINK

We will walkthrough the important steps in working with TestLink

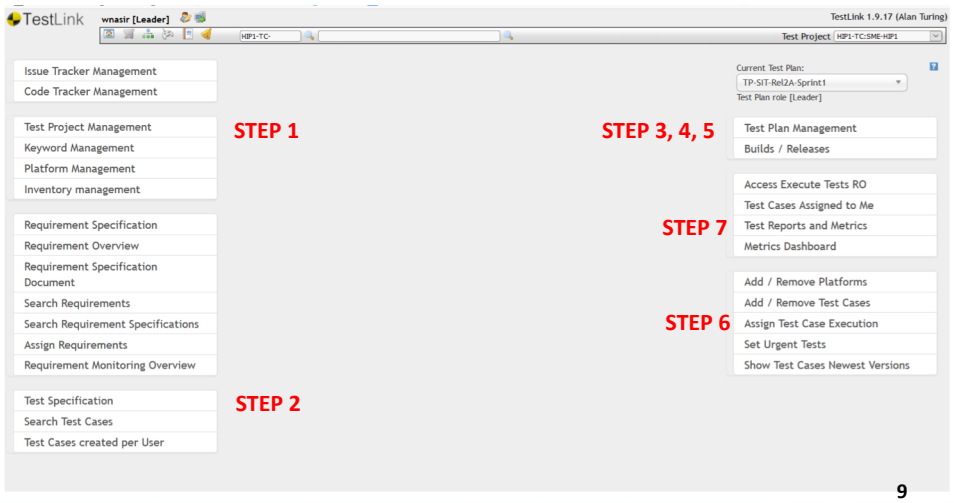
- **STEP 1** - Create a Project
- **STEP 2** - Create Test Cases (and Test Suites)
- **STEP 3** - Create Test Plan
- **STEP 4** - Specify Build of the Software Under Test (SUT)
- **STEP 5** - Add Test Cases to the Test Plan
- **STEP 6** - Assign Test Cases to Test Engineers
- **STEP 7** - Execute Test Cases (Test Engineers)
- **STEP 8** - See Reports and Charts

PART II

CREATING A PROJECT, TEST SUITE, AND TEST CASES

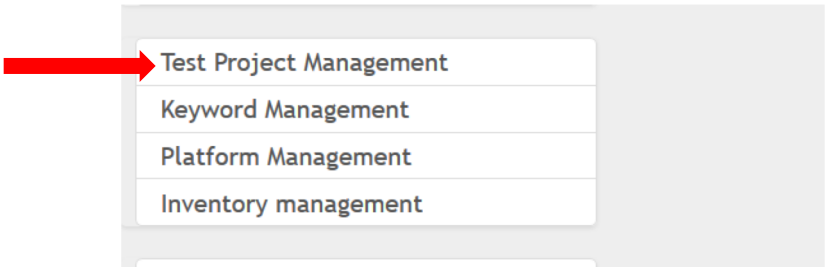
TESTLINK MAIN SCREEN

After successful login, here is how the main page of TestLink looks like:



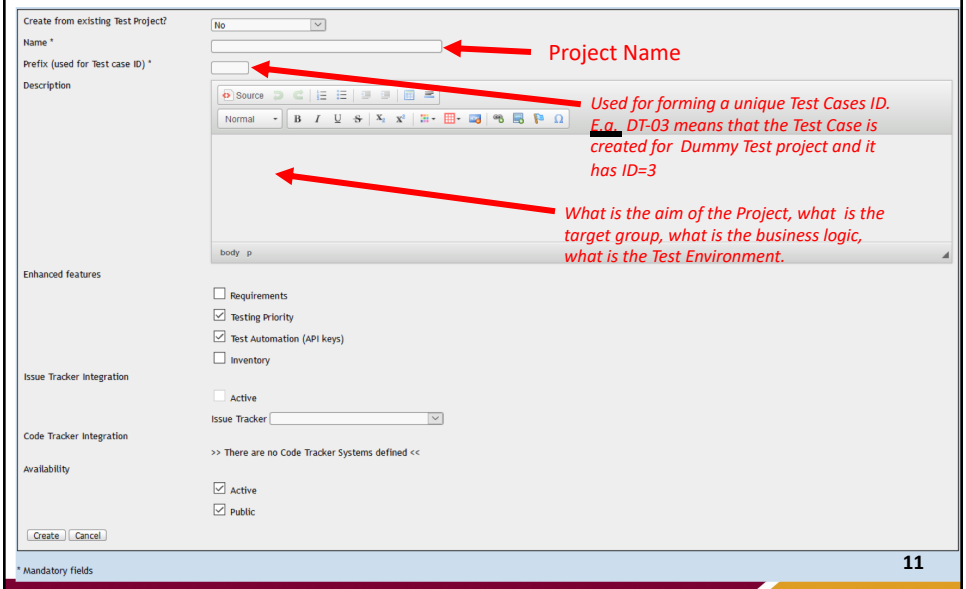
STEP 1: CREATE A PROJECT

*To create a project go to the **Test Project Management** section*



Premier Digital Tech University™

STEP 1. CREATE A PROJECT



11

STEP 1. CREATE A PROJECT

Enhanced features:

Requirements feature – we may specify requirements and see if they are well-covered by Test Cases

Testing priority – we may assign priority to Test Cases (high, medium, low)

Test Automation – we may specify whether the test should be performed manually or automatically

Inventory – to tell the truth, I didn't understand what it means

Issue Tracker:

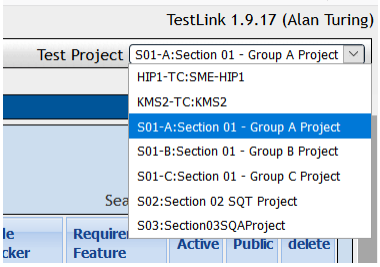
- we may link to issue tracker such as Redmine or JIRA
- Link test execution result with bugs/defects in the issue tracker

Premier Digital Tech University™

12

STEP 1. CREATE A PROJECT

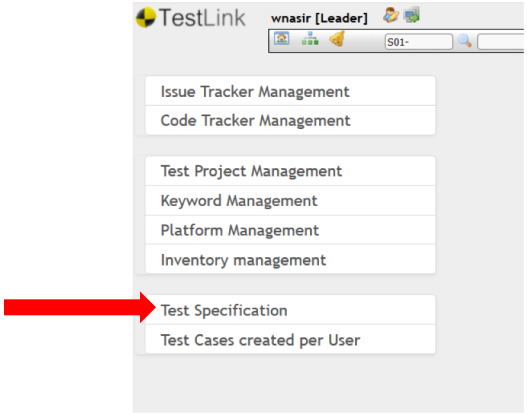
You can now select the Project that you will work on:



Premier Digital Tech University™

STEP 2. CREATE TEST CASES

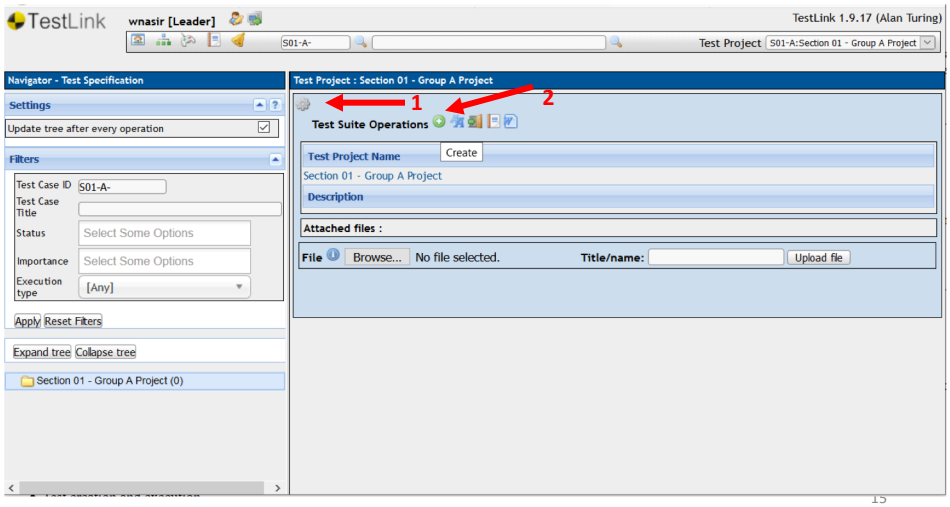
To create the Test Cases for the selected project, select Test Specification from the main page:



Premier Digital Tech University™

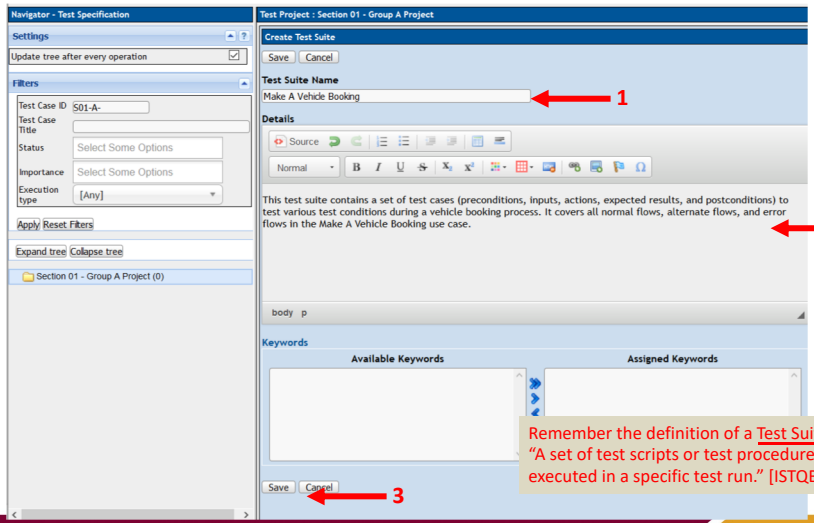
STEP 2. CREATE TEST CASES

Test Suites are created using the following buttons :



STEP 2. CREATE TEST CASES

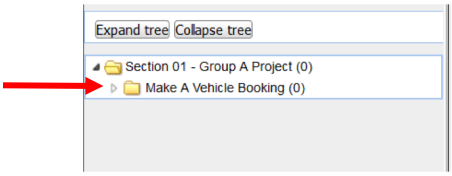
Test Suite details can be specified in the following form:



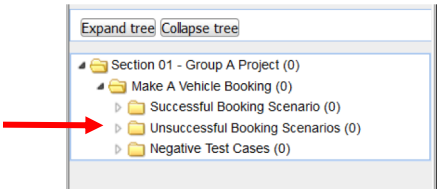
Remember the definition of a Test Suite?
"A set of test scripts or test procedures to be executed in a specific test run." [ISTQB]

STEP 2. CREATE TEST CASES

The newly created Test Suite will be available on the left menu:



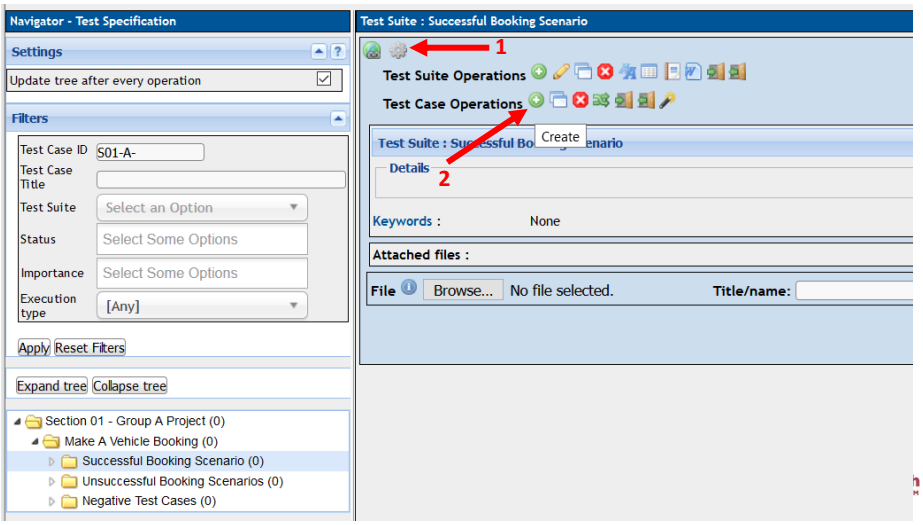
For better management, we may also add multiple sub Test Suites:



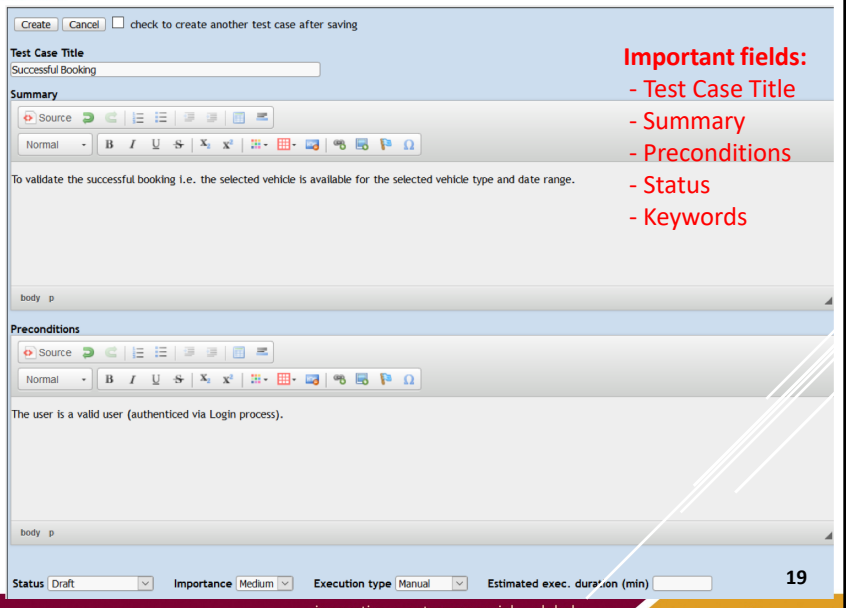
Premier Digital Tech University™

STEP 2. CREATE TEST CASES

How to create a Test Case:



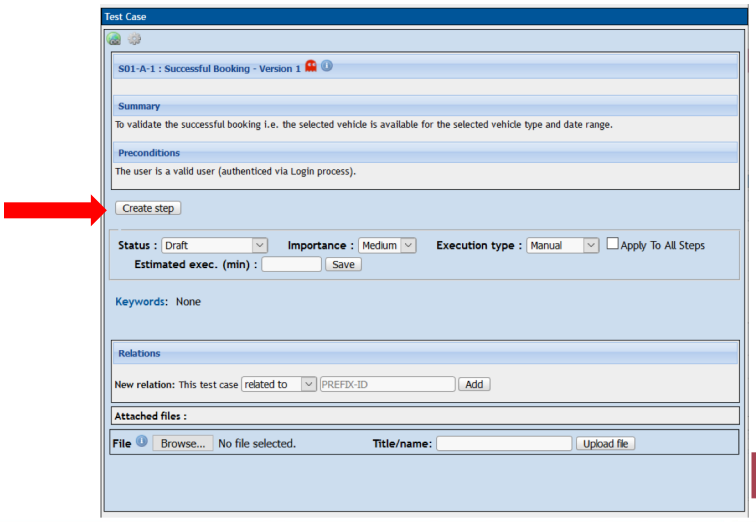
STEP 2. CREATE TEST CASES



Important fields:
- Test Case Title
- Summary
- Preconditions
- Status
- Keywords

STEP 2. CREATE TEST CASES

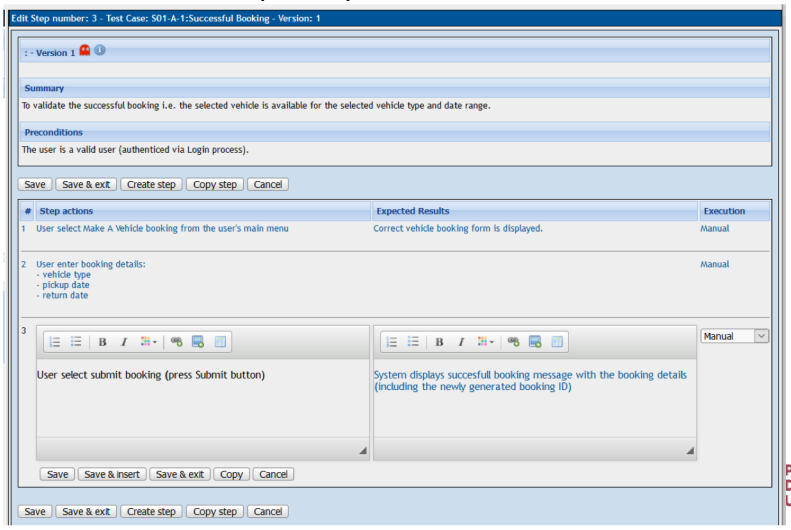
Next, we need to add steps to our Test Case:



Premier Digital Tech University™

STEP 2. CREATE TEST CASES

We could have multiple steps in a Test Case:



Premier Digital Tech University™

21

PART III

CREATING A TEST PLAN, SPECIFY BUILD, AND ADDING THE TEST CASES TO THE TEST PLAN

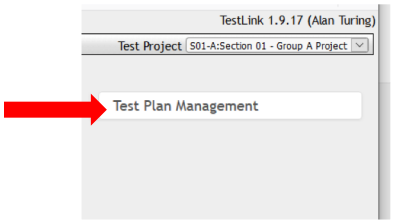
Premier Digital Tech University™

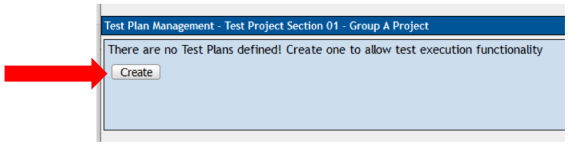
22

STEP 3. CREATE TEST PLAN

We cannot execute the Test Suites or Test Cases if we don't create a Test Plan and specify the Test Build.

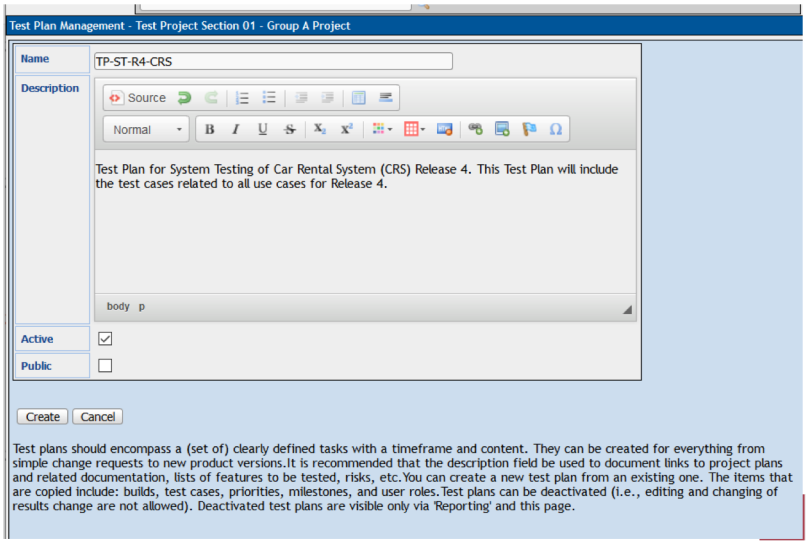
Therefore, we need to create a Test Plan:





Premier Digital Tech University™

STEP 3. CREATE TEST PLAN

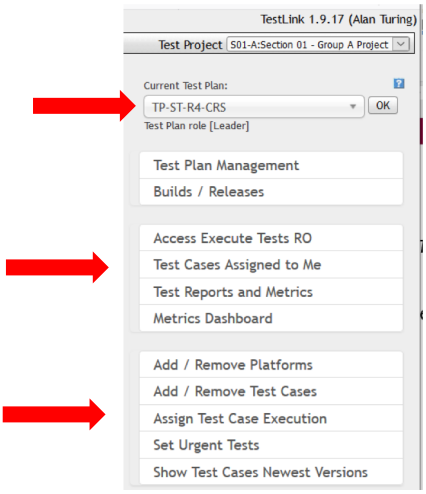


Premier Digital Tech University™

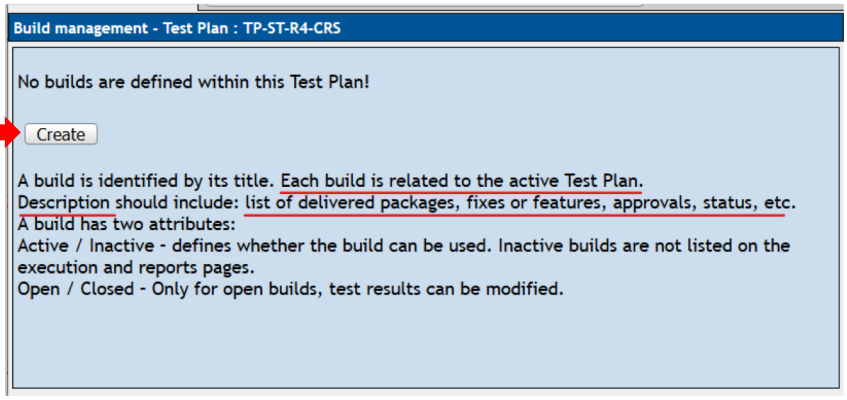
STEP 3. CREATE TEST PLAN

Current Test Plan will appear in the top right browser corner

After you've added a Test Plan, menu appear.

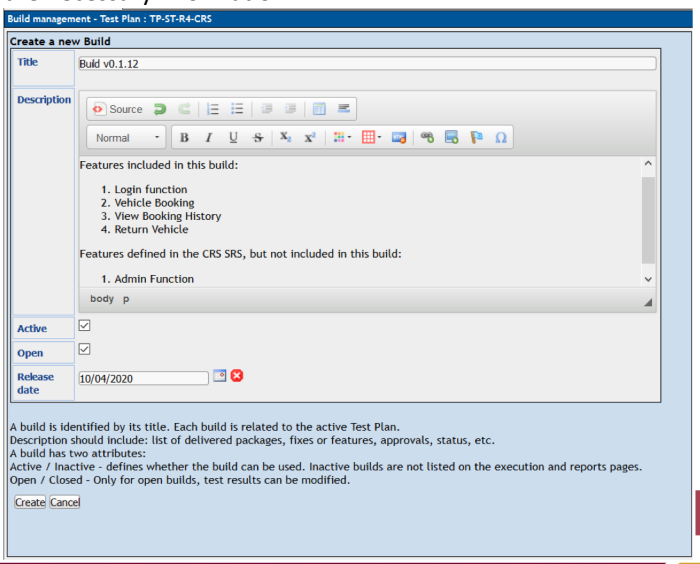


STEP 4. SPECIFY BUILD



STEP 4. SPECIFY BUILD

Fill in the necessary information:



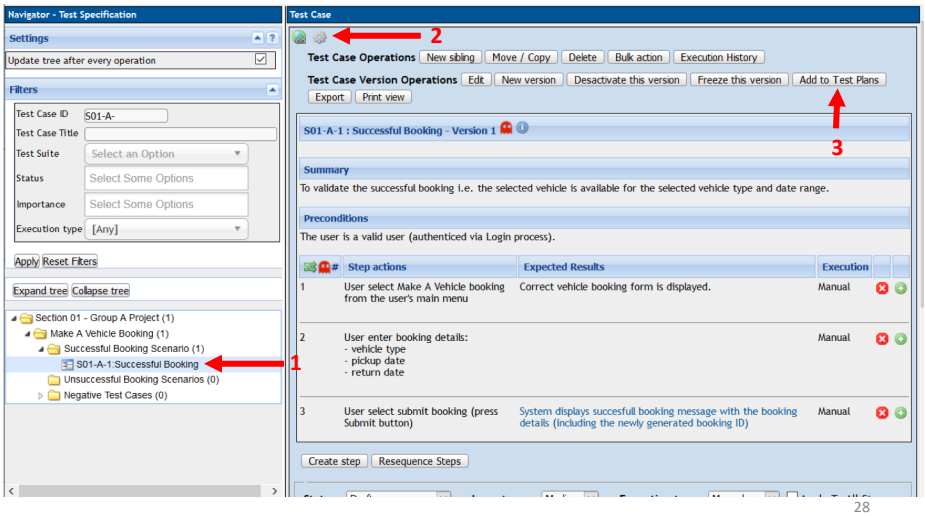
Premier Digital Tech University™

27

innovative • entrepreneurial • global www.utm.my

STEP 5. ADD TEST CASES TO A PLAN

Steps: 1. Select the Test Cases, Click , Click Add to Test Plans



28

innovative • entrepreneurial • global www.utm.my

STEP 5. ADD TEST CASES TO A PLAN

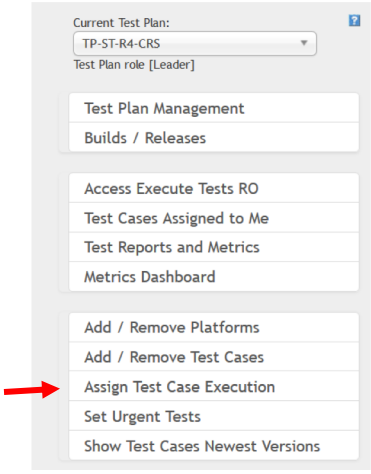
Then, you can choose what Test Plan you want to add the selected TC to:



Premier Digital Tech University™

STEP 5. ADD TEST CASES TO A PLAN

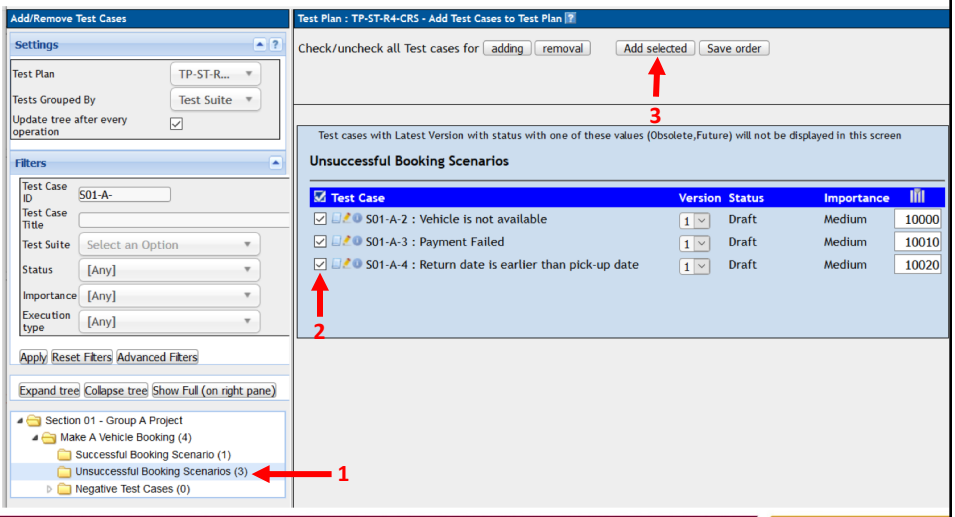
Alternatively, Test Case can be assigned to Test Plans from Desktop:



Premier Digital Tech University™

STEP 5. ADD TEST CASES TO A PLAN

Multiple selection is possible:



PART IV

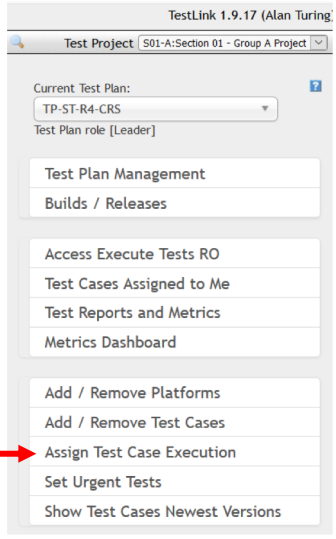
ASSIGNING TEST CASE EXECUTION, EXECUTING TEST CASES, AND GENERATING TEST REPORTS

Premier Digital Tech University™

32

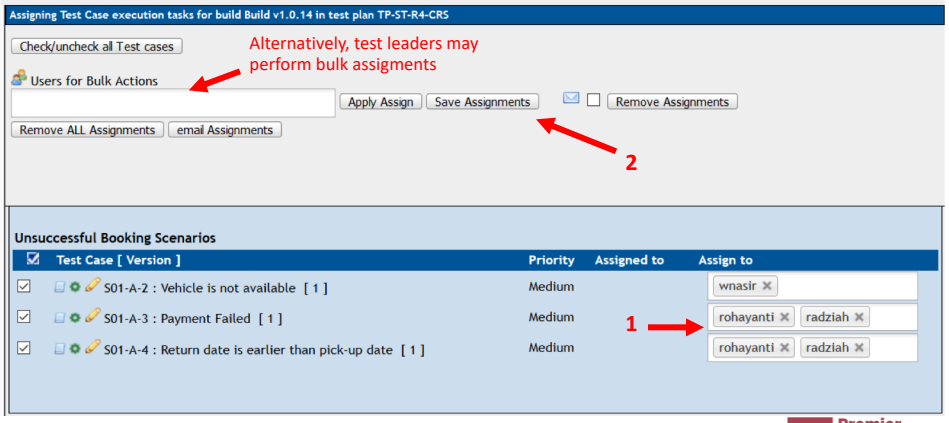
STEP 6. ASSIGN TC EXECUTION TO TESTERS

Test Case can't be executed before its assigned to a user (Tester)



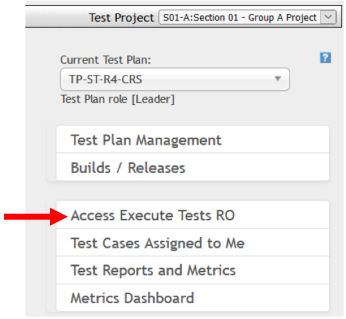
STEP 6. ASSIGN TC EXECUTION TO TESTERS

This page allows test leaders to assign users to particul tests within the Test Plan.



STEP 7. EXECUTE TESTS

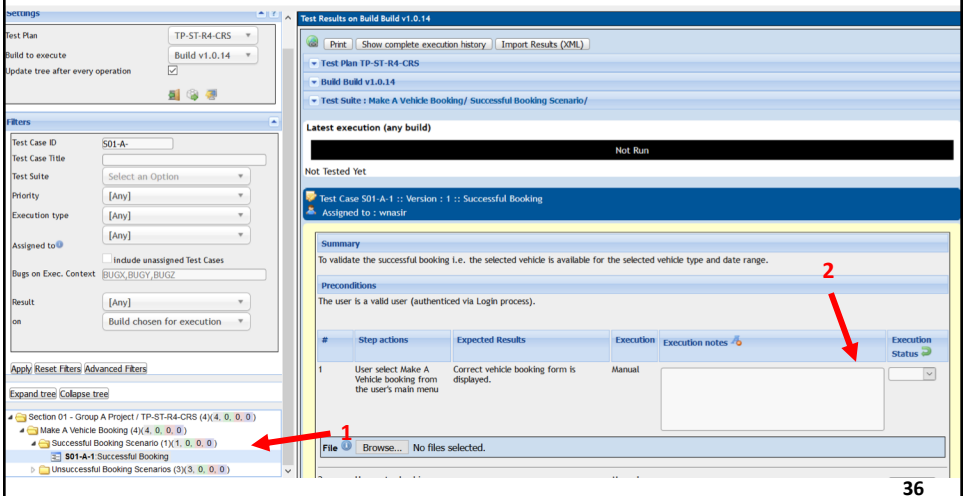
To start executing tests, Testers (or Test Engineers) should go to Test Execution section:



Premier Digital Tech University™

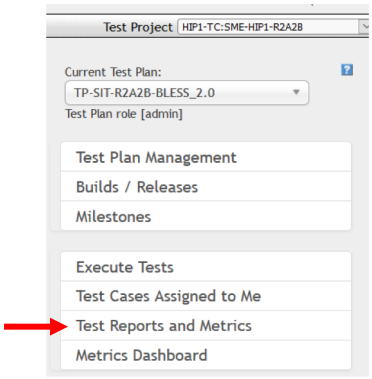
STEP 7. EXECUTE TESTS

In this section the user can register the results of the tests and track execution progress:



STEP 8. SEE REPORTS AND CHARTS

After the test case execution is finished, we may see the results by accessing the Test Reports section.

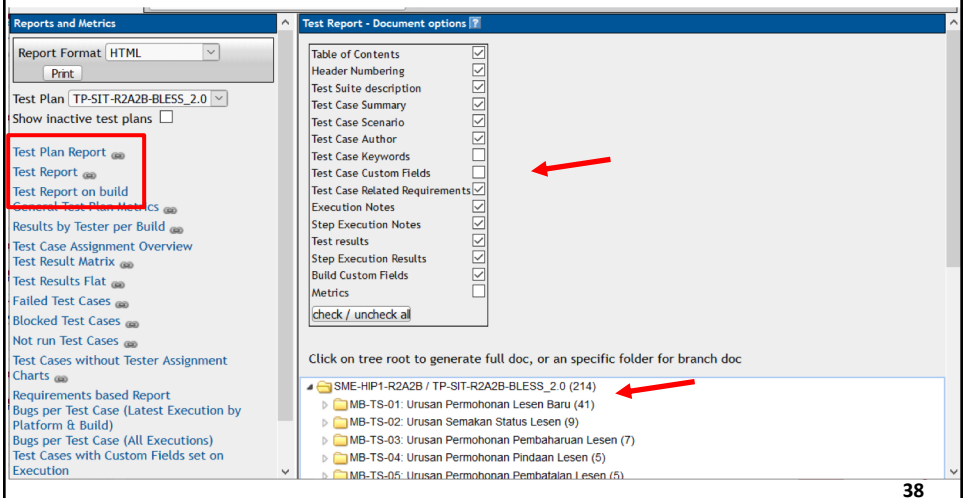


Premier Digital Tech University™

37

STEP 8. SEE REPORTS AND CHARTS

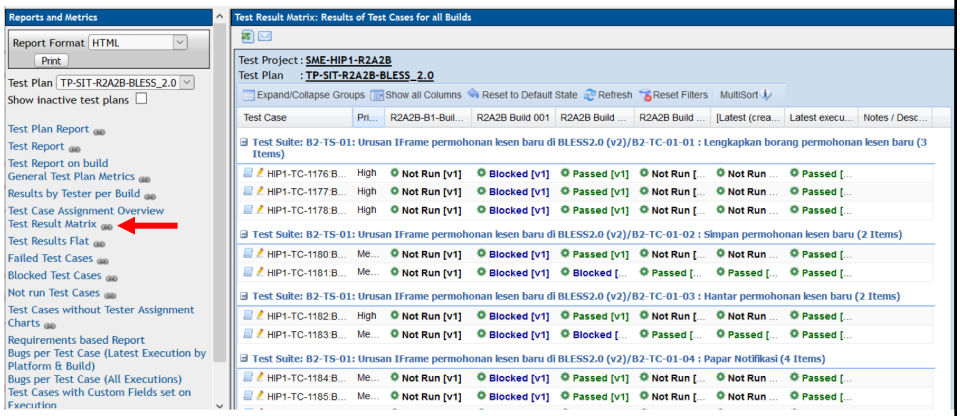
For Test Plan Report, Test Report, and Test Report on build, we may select the contents and the root Test Suite:



38

STEP 8. SEE REPORTS AND CHARTS

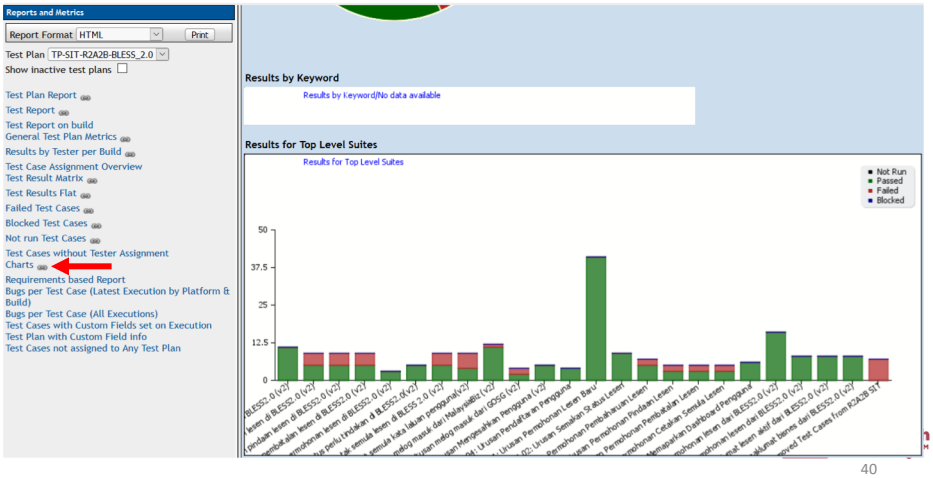
This is an example of the Test Result Matrix report. We can see the Not Run, Blocked, and Passed Test Cases executed for each Build.

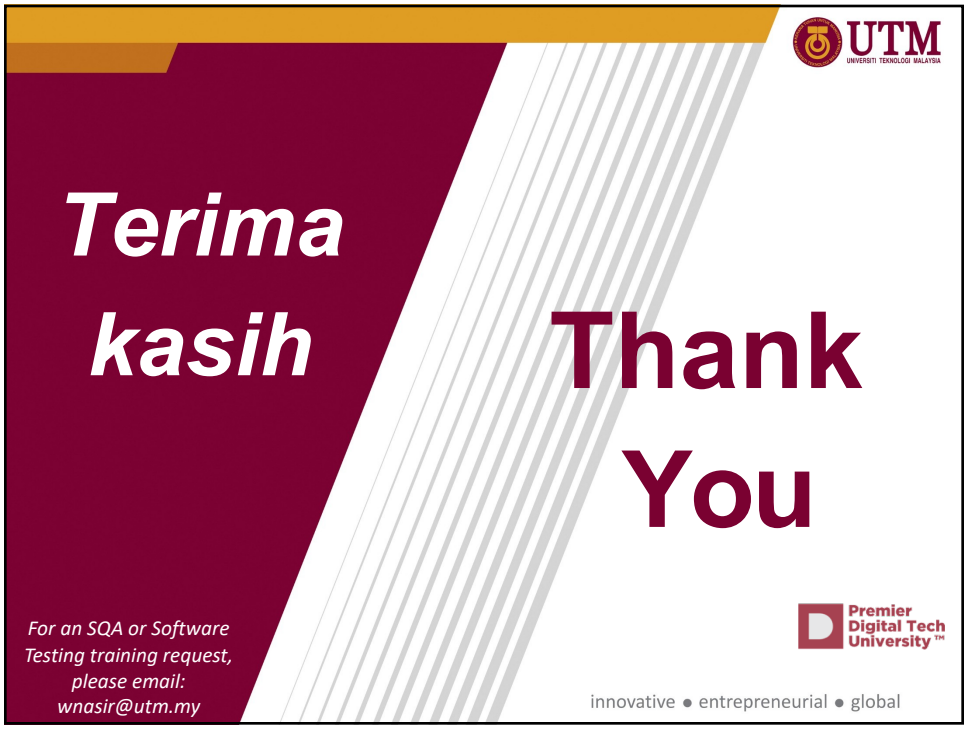


Premier Digital Tech University™

STEP 8. SEE REPORTS AND CHARTS

This is an example of the generated Chart. We can see the Not Run, Passed, Failed, and Blocked Test Cases.




UTM
UNIVERSITI TEKNOLOGI MALAYSIA
Terima kasih
Thank You
For an SQA or Software Testing training request, please email: wnasir@utm.my
Premier Digital Tech University™
innovative ● entrepreneurial ● global